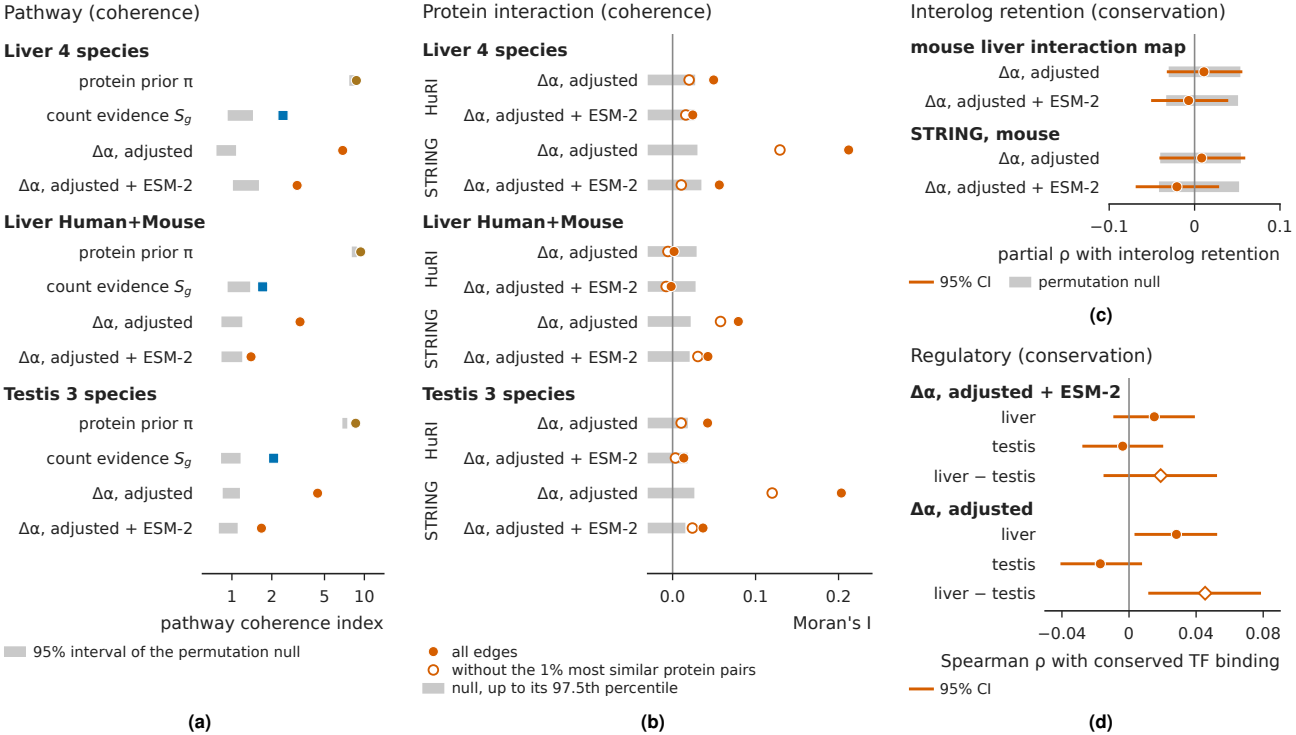

Pathway (coherence)
Liver 4 species
protein prior π
count evidence S_g
Δα, adjusted
Δα, adjusted + ESM-2
Liver Human+Mouse
protein prior π
count evidence S_g
Δα, adjusted
Δα, adjusted + ESM-2
Testis 3 species
protein prior π
count evidence S_g
Δα, adjusted
Δα, adjusted + ESM-2
1
2
5
10
pathway coherence index
95% interval of the permutation null
(a)
Protein interaction (coherence)
Liver 4 species
HuRI
Δα, adjusted
Δα, adjusted + ESM-2
STRING
Δα, adjusted
Δα, adjusted + ESM-2
Liver Human+Mouse
HuRI
Δα, adjusted
Δα, adjusted + ESM-2
STRING
Δα, adjusted
Δα, adjusted + ESM-2
Testis 3 species
HuRI
Δα, adjusted
Δα, adjusted + ESM-2
STRING
Δα, adjusted
Δα, adjusted + ESM-2
0.0
0.1
0.2
Moran's I
all edges
without the 1% most similar protein pairs
null, up to its 97.5th percentile
(b)
Interolog retention (conservation)
mouse liver interaction map
Δα, adjusted
Δα, adjusted + ESM-2
STRING, mouse
Δα, adjusted
Δα, adjusted + ESM-2
−0.1
0
0.1
partial ρ with interolog retention
95% CI
permutation null
(c)
Regulatory (conservation)
Δα, adjusted + ESM-2
liver
testis
liver − testis
Δα, adjusted
liver
testis
liver − testis
−0.04
0
0.04
0.08
Spearman ρ with conserved TF binding
95% CI
(d)